

Alexander Sanchez Cyril

Software Engineer | Full-Stack Development | AI Systems & Backend Engineering

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of hands-on experience building full-stack, backend, and AI-powered applications through independent software development projects. Experienced in designing and implementing scalable APIs, authentication systems, real-time communication systems, distributed architectures, and intelligent agent workflows using Python, TypeScript, C#, Java, Go, FastAPI, Django, ASP.NET Core, Next.js, LangGraph, LangChain, and LlamaIndex. Built and deployed SaaS platforms, AI applications, infrastructure monitoring tools, and developer productivity systems using event-driven architectures, caching systems, and cloud deployment practices. Strong foundation in software engineering principles, system design, cloud technologies, and modern development practices with a focus on scalability, maintainability, and reliability.

KEY ACHIEVEMENTS

- Built scalable APIs, authentication services, messaging systems, and database-driven applications using Python, TypeScript, C#, Java, Go, Django, FastAPI, ASP.NET Core, Fastify, Gin, and Next.js
- Developed SaaS platforms, AI-powered applications, infrastructure monitoring tools, and developer productivity systems from architecture through deployment
- Implemented distributed system concepts including caching, asynchronous messaging, service communication, event-driven workflows, and session management using Redis, RabbitMQ, Kafka, and REST APIs
- Designed and deployed cloud-hosted applications using Docker, CI/CD pipelines, Vercel, Render, Neon, and Infrastructure as Code with Terraform

TECHNICAL SKILLS

Languages

Python · JavaScript · TypeScript · C# · Java · Go

Backend Frameworks & APIs

ASP.NET Core · Spring Boot · Gin · Django · FastAPI · Express · Fastify

Frontend

React.js · Next.js · Tailwind CSS · HTML5 · CSS3 · Bootstrap

AI & Agentic Systems

LangGraph · LangChain · LangSmith · LlamaIndex · RAG · MCP · Vector Databases · Multi-Agent Systems · Prompt Engineering

Databases

PostgreSQL · MongoDB · MySQL · SQLite · Vector Databases · SQL Design & Optimization

Cloud & DevOps

Docker · Docker Compose · GitHub Actions · CI/CD · Vercel · Render · Neon · API Monitoring Systems

Infrastructure as Code

Terraform · Infrastructure as Code (IaC) concepts · Cloud architecture fundamentals

Backend & Distributed Systems Engineering

System Design · Clean Architecture · Microservices · Event-Driven Systems · REST APIs · API Design · JWT Authentication · RabbitMQ · Apache Kafka · Redis

Tools

Git · Linux basics · Postman · Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

SELF-EMPLOYED SOFTWARE ENGINEER | April 2023 – Present

Architected, developed, and deployed multiple full-stack, backend, AI-powered, and infrastructure-focused applications. Responsible for end-to-end software engineering including system architecture, backend development, frontend implementation, database design, API development, cloud deployment, and ongoing maintenance across multiple independent software systems.

EL-Exousia Weather Intel — Agentic Weather Intelligence Platform

Stack: *FastAPI · React · LangGraph · LangChain · LangSmith · LlamaIndex · MCP*

- Built an AI-powered weather intelligence system using multi-agent orchestration with LangGraph and tool-calling LLM workflows
- Designed a Retrieval-Augmented Generation (RAG) pipeline with vector memory for contextual weather reasoning and historical analysis
- Integrated real-time weather API services into an agentic reasoning pipeline for dynamic forecast generation
- Implemented modular agent workflows enabling multi-step decision-making and structured response generation

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AxioQuan — E-Learning & Marketplace Platform

Stack: *Java · C# · Fastify (TypeScript) · Next.js*

- Designed and developed a full-stack e-learning and digital marketplace platform with course creation, quizzes, and paid content delivery
- Implemented role-based access control system supporting students, instructors, and administrators
- Built real-time messaging system enabling communication between students and instructors
- Developed payment integration flow for course enrollment and content access control
- Engineered backend APIs and database schemas for scalable content management and delivery

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PingRobot — URL Monitoring & Wake-Up Service

Stack: *Next.js (Full-Stack) · TypeScript*

- Developed a cloud service monitoring system that prevents cold starts by periodically pinging deployed applications on free-tier platforms
- Implemented intelligent health-check system with configurable timeout handling and retry logic for service recovery detection
- Designed event-driven monitoring flow distinguishing between cold-start latency and active service responses
- Built secure API key authentication system and exposed REST API endpoints for external integration and testing
- Implemented user authentication using OAuth (Google/GitHub) and dashboard-based service management interface

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SchemaLence — PostgreSQL Schema Explorer & Database Analysis Tool

Stack: *Python · Tkinter · TTKBootstrap · Psycopg2 · Pandas*

- Built a desktop database analysis tool for PostgreSQL schema exploration, documentation, and inspection

- Implemented automated schema discovery including tables, relations, constraints, indexes, triggers, and ENUM definitions
- Developed interactive database browsing system with sortable table views and live record inspection
- Added export functionality for CSV, clipboard integration, and automated schema documentation generation
- Designed a desktop UI for structured database navigation and developer productivity workflows

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Universal Time — World Clock & Timezone API Platform

Stack: *TypeScript · Express.js · Next.js*

- Developed REST API for global time synchronization and timezone conversion services
- Built frontend interface for real-time analog and digital clock visualization
- Integrated API-driven architecture for real-time data rendering and synchronization across clients

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LEADERSHIP & ACADEMIC PROJECT EXPERIENCE

- Led university software development teams coordinating planning, task distribution, and system design decisions
- Acted as technical lead in Agile-style environments managing sprint planning and feature delivery
- Facilitated collaboration through code reviews, debugging support, and technical guidance
- Applied software engineering practices including Git workflows, testing, and system design
- Delivered successful team-based software projects from requirements to final deployment.

EDUCATION

Bachelor of Science in Software Development

Brigham Young University – Idaho

2026

Associate of Applied Science in Software Development

Brigham Young University – Idaho

2026

CERTIFICATIONS

Software Development · Brigham Young University – Idaho

2026

Web Development · Brigham Young University – Idaho

2025

Web & Computer Programming · Brigham Young University – Idaho

2024